

ONBRD-03: Deploying ActiveGate

Series: ONBRD — Dynatrace Onboarding | **Notebook:** 3 of 10 | **Created:** December 2025 | **Last Updated:** 08/03/2026

Your Network Gateway to Dynatrace

ActiveGate is a lightweight component that routes traffic between your infrastructure and Dynatrace. This notebook covers when you need ActiveGate, how many to deploy, where to place them, and installation steps - including comprehensive Kubernetes deployment options.

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Prerequisites

- Admin access to Dynatrace environment

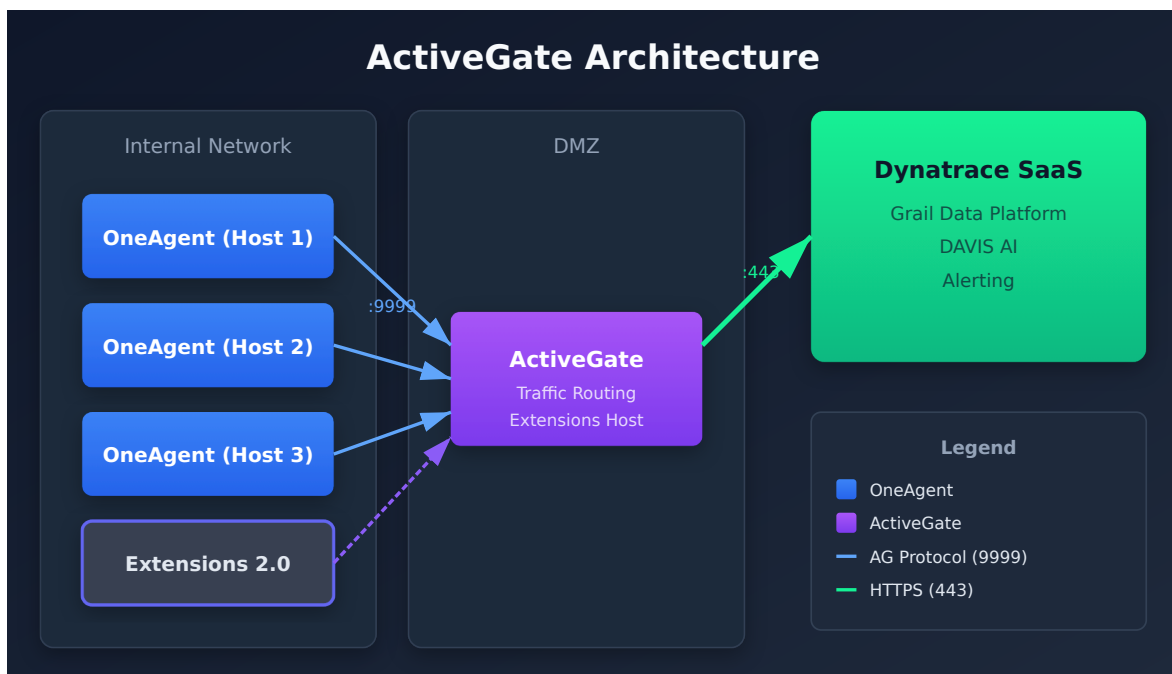
- Network architecture diagram (know your zones)
- Server for ActiveGate (Linux, Windows, or Kubernetes cluster)
- Outbound HTTPS (443) to Dynatrace SaaS

1. What is ActiveGate?

ActiveGate is a proxy and routing component that connects your environment to Dynatrace.

Component	Purpose
Environment ActiveGate	Routes OneAgent traffic, runs extensions, executes synthetics
Cluster ActiveGate	Dynatrace Managed only - cluster communication

Note: This notebook covers Environment ActiveGate for Dynatrace SaaS.



ActiveGate Capabilities

Capability	Description
Traffic Routing	Proxies OneAgent data to Dynatrace

Capability	Description
Extensions 2.0	Runs Python-based extensions locally
Synthetic Monitoring	Private synthetic locations
Cloud Integrations	AWS, Azure, GCP metric collection (or use Clouds app for direct connections)
Kubernetes API	Cluster monitoring via API
Log Ingest	Generic log ingest endpoint

OneAgent Attribute Enrichment (1.331+): OneAgent can enrich all telemetry (metrics, spans, logs, events) with primary fields (`dt.security_context` , `dt.cost.costcenter`) and primary tags (`primary_tags.environment` , `primary_tags.team`) at the source. More efficient than auto-tags — feeds directly into OpenPipeline routing, bucket assignment, and Grail permissions. Configure via `oneagentctl --set-host-tag` or `--set-host-tag` at install time. See [docs](#).

Dynatrace Version Support Policy

Component	Standard Support	Enterprise Support
OneAgent	9 months	12 months
ActiveGate	9 months	12 months
Dynatrace Operator	Independent release cycle — check release notes	

Technology Support Tiers

Tier	Meaning
Full Support	All monitoring features maintained, bugs fixed, enhancements delivered
Early Adopter	Newly introduced — functional but may lack full feature parity; feedback welcome

Tier	Meaning
End of Life (EOL)	No updates, fixes, or enhancements; monitoring may still work but issues won't be addressed

Third-Party Technology EOL Policy

Dynatrace continues to support monitoring a third-party technology for **6 months beyond the vendor's end-of-life date**. End-of-support announcements are published 6 months in advance.

Reference: [Technology Support Model](#) | [End-of-Support Announcements](#) | [Support Policy](#)

2. When Do You Need ActiveGate?

Required Scenarios

Scenario	Why ActiveGate is Required
Network-restricted hosts	OneAgents can't reach internet directly
Private synthetic monitors	Test internal applications
Extensions 2.0	Custom data sources (SNMP, databases, etc.)
AWS / Azure / GCP monitoring	See Clouds app status below — AG still required for GCP and for environments without Clouds app
VMware monitoring	vCenter integration
Kubernetes full-stack	Cluster API access for events, metrics

Update — Clouds App (status as of 2026-05): The **Clouds app** supports direct cloud connections **without ActiveGate** — but coverage varies per cloud:

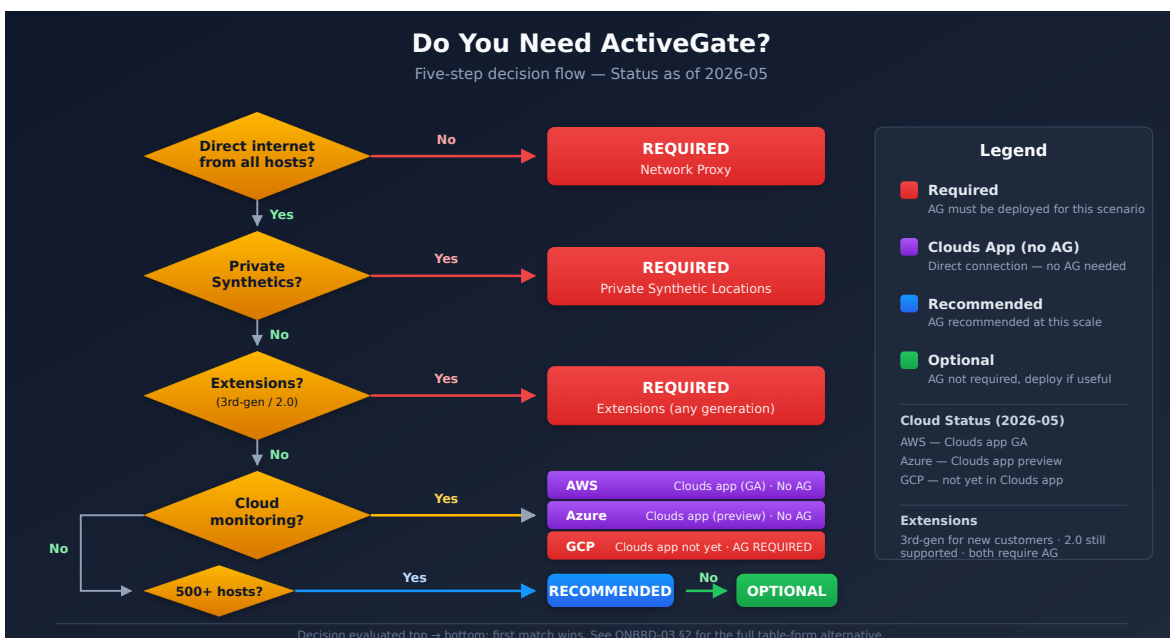
Cloud	Clouds-app Status	AG Required?
AWS	GA — direct connection supported	No (when using Clouds app)
Azure	Preview — direct connection supported	No (when using Clouds app preview)
GCP	Not yet available in Clouds app	Yes — AG-based polling

ActiveGate is still required for **Extensions 2.0**, **private synthetic locations**, **GCP monitoring**, and any environment without Clouds app access. See [Clouds app documentation](#) and the **CLOUD series** for per-provider deep dives.

Optional but Recommended

Scenario	Benefit
Large deployments (500+ hosts)	Reduces outbound connections
Multiple network zones	Centralized routing per zone
Bandwidth optimization	Compresses and batches data
Security compliance	Single egress point for audit

Decision Tree



3. How Many ActiveGates?

Sizing Guidelines

FAQ-10 owns the ActiveGate sizing decision. *FAQ-10: How do I size and scale ActiveGates?* carries the current published per-shape capacity tables, the 50% CPU / 80% memory headroom rule, the survivor-capacity HA math for a multi-node zone, and the `dt.sfm.active_gate.*` saturation signals to watch once the fleet is running. Treat the table below as an order-of-magnitude sanity check while you plan the deployment — size the actual machines against FAQ-10 and the official requirements page.

Dynatrace publishes estimated host counts per reference instance shape, with the guidance that a steady-state ActiveGate *"should not exceed 50% CPU and 80% memory"*:

Reference shape	Architecture	vCPU	RAM	Estimated hosts routed
c6i.large	x86-64	2	3.75 GiB	~800
c6i.xlarge	x86-64	4	7.5 GiB	~1,800
c6i.2xlarge	x86-64	8	15 GiB	~2,500
c7g.large	ARM64	2	3.75 GiB	~1,300
c7g.xlarge	ARM64	4	7.5 GiB	~2,700
c7g.2xlarge	ARM64	8	15 GiB	~5,500

The reference points are AWS instance shapes — for on-premises VMs, map by vCPU/RAM rather than looking for the instance name. "Estimated hosts routed" assumes typical per-host data volume; hosts pushing heavy log volume through the ActiveGate consume capacity faster. Note also that ARM64 routes materially more hosts per vCPU than x86-64 at the same shape, which is worth knowing *before* you standardize on an instance family.

Detailed Hardware Requirements

Minimum System Requirements

Component	Minimum	Recommended
CPU	2 cores (x64)	4+ cores
RAM	2 GB	4+ GB
Disk Space	2 GB	10+ GB (for logs/cache)
Disk Type	HDD	SSD (for extensions)
Network	100 Mbps	1 Gbps

Operating System Support

The routing/monitoring ActiveGate matrix changes every few sprints, so **the official requirements pages are the authority** — re-check them before you provision, not just when something breaks: [Linux ActiveGate hardware and system requirements](#) and [Windows ActiveGate hardware and system requirements](#).

Currently listed for routing/monitoring ActiveGates (verified 07/30/2026, ActiveGate 1.343):

OS	Supported Versions
Red Hat Enterprise Linux	8.10, 9.4, 9.6, 9.7, 9.8, 10.0, 10.1, 10.2
Oracle Linux	8.10, 9.7, 9.8, 10.1, 10.2
Rocky Linux	8.10, 9.7, 9.8, 10.1, 10.2
Ubuntu	16.04, 18.04, 20.04, 22.04, 24.04, 26.04 (x86-64); 20.04, 22.04, 24.04, 26.04 (ARM64 / s390)
Amazon Linux	2, 2023
SUSE Linux Enterprise Server	15.7
Windows Server	2016, 2019, 2022, 2025

Added in ActiveGate 1.343: Oracle Linux 9.8 and 10.2, Rocky Linux 9.8 and 10.2.
ActiveGate 1.343 was published 07/15/2026 with a staged fleet rollout from 07/28/2026 —

the ActiveGates already running in your environment carry whatever version they last updated to, so check the version of the specific ActiveGate before assuming a newly added OS is installable on it.

On CentOS and Debian: neither CentOS Linux, CentOS Stream, nor Debian appears on the current routing/monitoring ActiveGate matrix (verified 07/30/2026). Earlier revisions of this notebook combined "RHEL/CentOS" into a single row, which reads as an endorsement of CentOS that the requirements page does not make — and CentOS Linux is end-of-life upstream regardless of what Dynatrace supports. If you are planning ActiveGates on CentOS Stream or Debian, check the requirements page directly before provisioning rather than inferring support from the RHEL row.

Support Ending — Do Not Provision New ActiveGates on These

Dynatrace de-supports an OS version *"at least 6 months after its end of life (EOL), to give you enough time to upgrade your environment."* The announced removals below all fall inside the next six months:

OS version	ActiveGate support ends
Red Hat Enterprise Linux 9.4	11/01/2026
Ubuntu 16.04	11/01/2026
Red Hat Enterprise Linux 9.7, 10.1	12/01/2026
Oracle Linux 9.7, 10.1	12/01/2026
Rocky Linux 9.7, 10.1	12/01/2026
Amazon Linux 2	01/01/2027

Existing ActiveGates on these versions keep working right up to the date — nothing stops the moment the announcement lands. What ends is Dynatrace's commitment to fix issues and ship new ActiveGate versions for that OS. Two practical consequences for an onboarding plan:

- **Do not provision new ActiveGates on any version in this table.** Amazon Linux 2 and RHEL 9.4 were the obvious defaults a year ago and are now the wrong starting point — pick Amazon Linux 2023, RHEL 9.8/10.2, Oracle Linux 9.8/10.2, or Rocky Linux 9.8/10.2 instead.

- **Fold the OS upgrade into the next ActiveGate update window** rather than scheduling it as separate work. The update mechanics, auto-update behavior, and how to schedule those windows are covered in *FAQ-05: Managing ActiveGate updates on Dynatrace SaaS*.

Reference: [End-of-support announcements \(DT docs\)](#) — the dates and the 6-month policy statement above are quoted from this page, verified 07/30/2026.

Memory Sizing by Workload

Workload Type	Base Memory	Additional per Feature
Routing Only	2 GB	-
+ Extensions 2.0	2 GB	+1-2 GB per active extension
+ Synthetic	2 GB	+1 GB per 10 monitors
+ Cloud Integration	2 GB	+512 MB per cloud account
+ Log Ingest	2 GB	+2 GB for high volume

Disk Space Sizing

Purpose	Space Required
Base Installation	1.5 GB
Log Files (default)	1-5 GB (rotated)
Extension Cache	500 MB - 2 GB
Synthetic Cache	500 MB - 1 GB
Temporary Files	1 GB
Recommended Total	10-20 GB

Best Practice: Mount `/opt/dynatrace` on a dedicated partition with at least 20 GB for production deployments.

High Availability

For production environments, deploy **at least 2 ActiveGates per network zone**:

Deployment	ActiveGates	Purpose
Minimum HA	2 per zone	Failover capability
Recommended	2-3 per zone	Load distribution + failover
Large scale	N+1 per zone	Capacity + redundancy

Key Point: OneAgents automatically load-balance across available ActiveGates and failover if one becomes unavailable.

Example Deployment

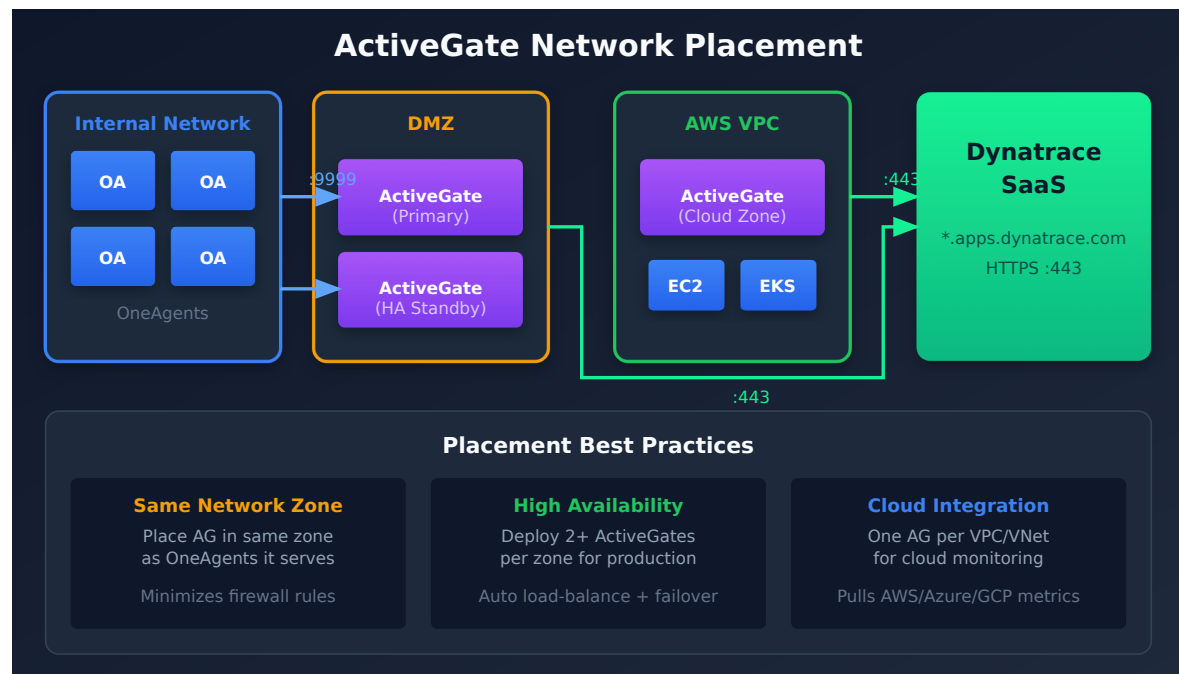
Network Zone	Hosts	ActiveGates	Shape class per AG	vCPU each	RAM each
Production DMZ	200	2	c6i.large (~800 hosts)	2	3.75 GiB
Production Internal	800	2	c6i.xlarge (~1,800 hosts)	4	7.5 GiB
Development	150	1	c6i.large (~800 hosts)	2	3.75 GiB
Total	1,150	5		14 vCPU	26.25 GiB

Note why Production Internal is sized at c6i.xlarge for only 800 hosts across two nodes: each node must be able to carry the **whole** zone when its partner is down, so the survivor's capacity — not the average load — sets the shape. FAQ-10 works that survivor-capacity math out properly, including for zones larger than two nodes.

4. Where to Deploy?

Network Zone Strategy

Deploy ActiveGates based on network segmentation:



Placement Rules

Rule	Description
Same network zone	AG should be in same zone as OneAgents it serves
Outbound access	AG needs HTTPS to *.dynatrace.com
Inbound from agents	OneAgents connect to AG on port 9999
Low latency	Place AG close to monitored workloads

Cloud-Specific Guidance

Cloud	Recommendation
AWS	Deploy in management VPC or shared services subnet
Azure	Hub VNet with peering to spoke VNets

Cloud	Recommendation
GCP	Shared VPC host project
Kubernetes	Deploy as StatefulSet or standalone VM outside cluster

5. Generating Tokens

ActiveGate requires a PaaS token for installation.

Location: Settings → Integration → Platform as a Service

Or use the API token approach:

Required Token Scopes

Scope	API Name	Purpose
PaaS integration - Installer download	InstallerDownload	Download ActiveGate installer

Additional Scopes by Use Case

Use Case	Additional Scopes
AWS Monitoring	aws.supportedServicesRead
Azure Monitoring	azure.supportedServicesRead
Extensions 2.0	extensions.read , extensions.write
Synthetic Private Location	syntheticLocations.write

Token Naming Convention

Use descriptive names:

- prod-activegate-dmz
- aws-activegate-useast1
- k8s-activegate-cluster1

6. Installation Methods

Linux Installation

```
# Download the installer
wget -O Dynatrace-ActiveGate.sh \
  "https://{tenant-id}.live.dynatrace.com/api/v1/deployment/installer/gateway/unix/latest?Api-Token={paas-token}"

# Make executable and run
chmod +x Dynatrace-ActiveGate.sh
sudo ./Dynatrace-ActiveGate.sh
```

Windows Installation

```
# Download the installer
Invoke-WebRequest -Uri "https://{tenant-id}.live.dynatrace.com/api/v1/deployment/installer/gateway/windows/latest?Api-Token={paas-token}" -OutFile Dynatrace-ActiveGate.exe

# Run the installer
.\Dynatrace-ActiveGate.exe
```

Deprecation note (SaaS 1.343, July 2026): the classic ActiveGate deployment API used above (`/api/v1/deployment/installer/gateway/...` with a PaaS/ Api-Token) is **deprecated** in favor of the Latest Dynatrace deployment REST API, which supports **platform tokens** with fine-grained OAuth scopes and adds a REST endpoint for public component-image URIs. Its GA/enabled-by-default status arrives with the staged SaaS 1.343 tenant rollout (from mid-July 2026) — verify availability in your tenant. The classic endpoints continue to work during the deprecation period — plan new automation against the platform API and migrate existing scripts on your next maintenance touch.

Container Deployment (Docker/Podman)

```
docker run -d --name dynatrace-activegate \  
-e DT_TENANT="{tenant-id}" \  
-e DT_API_TOKEN="{paas-token}" \  
-p 9999:9999 \  
dynatrace/dynatrace-activegate:latest
```

Installation Parameters

Parameter	Purpose	Example
<code>--set-network-zone</code>	Assign to network zone	<code>--set-network-zone=dmz</code>
<code>--set-group</code>	Group for management	<code>--set-group=production</code>
<code>--enable-synthetic</code>	Enable synthetic capability	<code>--enable-synthetic</code>

6a. Kubernetes Deployment (Detailed)

Deploying ActiveGate in Kubernetes requires careful consideration of where, how, and when to use containerized ActiveGates vs. traditional VM deployments.

When to Deploy ActiveGate in Kubernetes

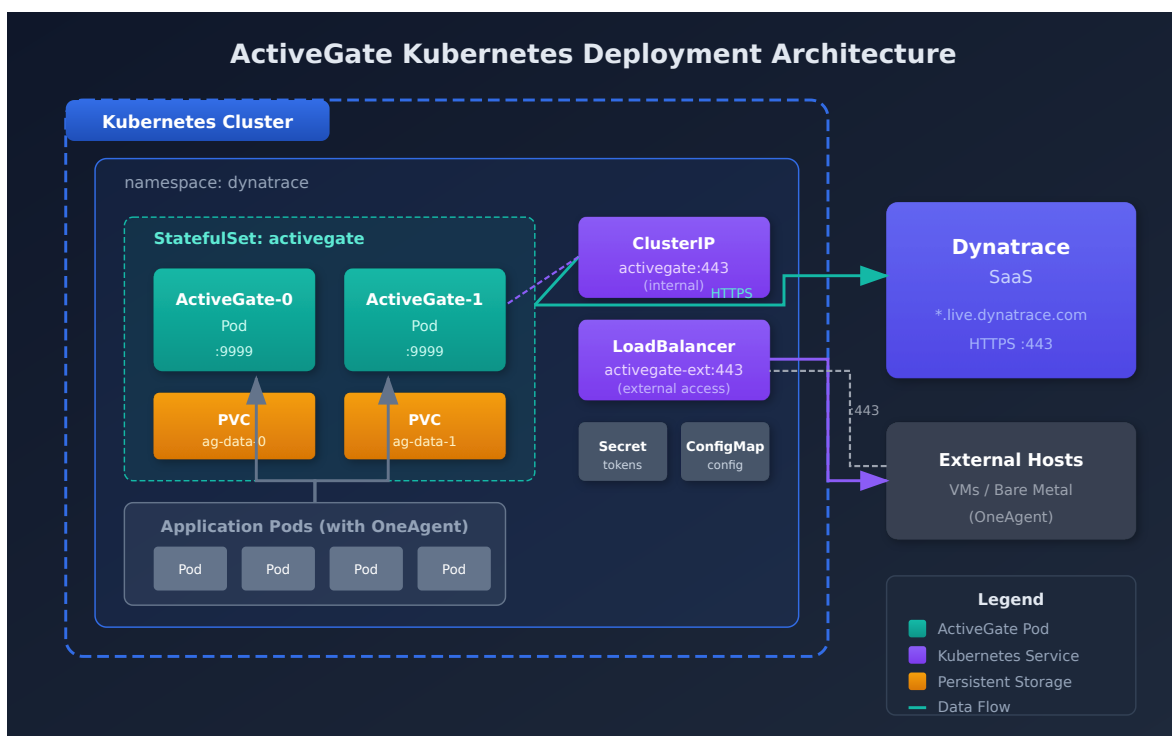
Scenario	Deploy in K8s?	Reasoning
Cluster-only monitoring	✅ Yes	Co-located with workloads, simplifies networking
Routing for in-cluster OneAgents	✅ Yes	Lower latency, no external hops
Extensions 2.0 for K8s resources	✅ Yes	Direct access to cluster APIs
Multi-cluster routing	⚠️ Consider	May need external AG for cross-cluster
Private synthetic monitoring	❌ Usually VM	Synthetic requires stable network, not pod restarts

Scenario	Deploy in K8s?	Reasoning
Routing for external hosts	✗ VM preferred	External hosts need stable, routable IP
Air-gapped environments	✗ VM preferred	Often need dedicated proxy tier

When NOT to Deploy in Kubernetes

Scenario	Why VM is Better
Routing for VMs/bare metal	VMs need stable IPs, not pod IPs
Private synthetic locations	Synthetic tests sensitive to pod restarts
Extensions polling external systems	External systems may not allow K8s egress IPs
Compliance requiring dedicated hosts	Some compliance requires dedicated hardware

Deployment Architecture



Kubernetes Hardware Requirements

Deployment Size	CPU Request	CPU Limit	Memory Request	Memory Limit
Small (≤500 agents)	500m	2000m	1Gi	2Gi
Medium (500-1500)	1000m	4000m	2Gi	4Gi
Large (1500-3000)	2000m	8000m	4Gi	8Gi
Extra Large (3000+)	4000m	16000m	8Gi	16Gi

Method 1: Dynatrace Operator (Recommended)

The Dynatrace Operator manages ActiveGate lifecycle automatically.

Important: Use `apiVersion: dynatrace.com/v1beta6` for new DynaKubes (`v1beta5` remains accepted). Operator **1.9.0** removed `v1beta3` from the CRD ("*Applying DynaKube resources using this version will fail*"), and Operator **1.10.0** (July 15, 2026) deprecates `v1beta4` — check `kubectl get dynakube -A -o jsonpath='{.items[*].apiVersion}'` before upgrading the Operator. `v1beta6` adds OTLP exporter configuration.


```
# dynakube.yaml - ActiveGate via Operator
apiVersion: dynatrace.com/v1beta5
kind: DynaKube
metadata:
  name: dynakube
  namespace: dynatrace
spec:
  apiUrl: https://{tenant-id}.live.dynatrace.com/api

  # ActiveGate configuration
  activeGate:
    # Capabilities to enable
    capabilities:
      - routing          # Route OneAgent traffic
      - kubernetes-monitoring # K8s API monitoring
      - dynatrace-api     # Proxy Dynatrace API calls

  # Resource sizing
  resources:
    requests:
      cpu: "500m"
      memory: "1Gi"
    limits:
      cpu: "2000m"
      memory: "2Gi"

  # Number of replicas (HA)
  replicas: 2

  # Tolerations for dedicated nodes
  tolerations:
    - key: "dedicated"
      operator: "Equal"
      value: "dynatrace"
      effect: "NoSchedule"

  # Node selector (optional)
  nodeSelector:
    node-type: monitoring

  # Custom properties
  customProperties:
    value: |
      [connectivity]
      networkZone=kubernetes

  # Labels
```

```
labels:  
  app.kubernetes.io/component: activegate
```

Install with Operator:

```
# Create namespace  
kubectl create namespace dynatrace  
  
# Create secret with tokens – apiToken (with PaaS scopes) is sufficient for  
the Operator;  
# the separate paasToken secret field is deprecated as of Operator 1.10.0  
(still accepted).  
# From Operator 1.10.0 a Platform Token is also accepted in place of the  
classic access token.  
kubectl -n dynatrace create secret generic dynakube \  
  --from-literal=apiToken=<your-api-token>  
  
# Install Dynatrace Operator via Helm  
helm repo add dynatrace https://raw.githubusercontent.com/Dynatrace/dynatrace-  
operator/main/config/helm/repos/stable  
helm repo update  
  
helm install dynatrace-operator dynatrace/dynatrace-operator \  
  --namespace dynatrace \  
  --set installCRD=true  
  
# Apply DynaKube configuration  
kubectl apply -f dynakube.yaml
```

Method 2: Helm Chart (Standalone)

For more control, use the standalone ActiveGate Helm chart:

```
# Add Dynatrace Helm repo
helm repo add dynatrace https://raw.githubusercontent.com/Dynatrace/dynatrace-operator/main/config/helm/repos/stable
helm repo update

# Create values file
cat > activegate-values.yaml <<< 'EOF'
apiUrl: "https://{tenant-id}.live.dynatrace.com/api"

activeGate:
  replicas: 2

  capabilities:
    - routing
    - kubernetes-monitoring

  resources:
    requests:
      cpu: 500m
      memory: 1Gi
    limits:
      cpu: 2
      memory: 2Gi

# Persistent storage for logs
persistence:
  enabled: true
  size: 10Gi
  storageClassName: gp3

# Service configuration
service:
  type: ClusterIP # or LoadBalancer for external access

# Environment variables
env:
  - name: DT_NETWORK_ZONE
    value: "kubernetes"

# Token from existing secret
existingSecret: dynatrace-tokens
EOF

# Install
helm install activegate dynatrace/dynatrace-activegate \
  --namespace dynatrace \
  --values activegate-values.yaml
```

Method 3: Raw Kubernetes Manifests

For complete control, use raw manifests:

```

# activegate-namespace.yaml
apiVersion: v1
kind: Namespace
metadata:
  name: dynatrace
---
# activegate-secret.yaml
apiVersion: v1
kind: Secret
metadata:
  name: dynatrace-tokens
  namespace: dynatrace
type: Opaque
stringData:
  apiToken: "<your-api-token>"
  paasToken: "<your-paas-token>"
---
# activegate-configmap.yaml
apiVersion: v1
kind: ConfigMap
metadata:
  name: activegate-config
  namespace: dynatrace
data:
  custom.properties: |
    [connectivity]
    networkZone=kubernetes

    [collector]
    MaxIncomingConnections=2000
---
# activegate-statefulset.yaml
apiVersion: apps/v1
kind: StatefulSet
metadata:
  name: activegate
  namespace: dynatrace
  labels:
    app: activegate
spec:
  serviceName: activegate
  replicas: 2
  selector:
    matchLabels:
      app: activegate
  template:
    metadata:

```

```
labels:
  app: activegate
spec:
  serviceAccountName: dynatrace-activegate
  containers:
    - name: activegate
      image: dynatrace/dynatrace-activegate:latest
      imagePullPolicy: Always
      env:
        - name: DT_TENANT
          value: "{tenant-id}"
        - name: DT_API_TOKEN
          valueFrom:
            secretKeyRef:
              name: dynatrace-tokens
              key: paasToken
        - name: DT_CAPABILITIES
          value: "routing,kubernetes_monitoring"
      ports:
        - containerPort: 9999
          name: ag-https
      resources:
        requests:
          cpu: "500m"
          memory: "1Gi"
        limits:
          cpu: "2000m"
          memory: "2Gi"
      volumeMounts:
        - name: config
          mountPath: /var/lib/dynatrace/gateway/config/custom.properties
          subPath: custom.properties
        - name: ag-data
          mountPath: /var/lib/dynatrace/gateway
      livenessProbe:
        httpGet:
          path: /rest/health
          port: 9999
          scheme: HTTPS
        initialDelaySeconds: 30
        periodSeconds: 15
      readinessProbe:
        httpGet:
          path: /rest/health
          port: 9999
          scheme: HTTPS
        initialDelaySeconds: 30
        periodSeconds: 5
```

```

    volumes:
      - name: config
        configMap:
          name: activegate-config
    volumeClaimTemplates:
      - metadata:
          name: ag-data
        spec:
          accessModes: ["ReadWriteOnce"]
          storageClassName: gp3
          resources:
            requests:
              storage: 10Gi
---
# activegate-service.yaml
apiVersion: v1
kind: Service
metadata:
  name: activegate
  namespace: dynatrace
spec:
  type: ClusterIP # Change to LoadBalancer for external access
  selector:
    app: activegate
  ports:
    - port: 443
      targetPort: 9999
      name: https
---
# Headless service for StatefulSet DNS
apiVersion: v1
kind: Service
metadata:
  name: activegate-headless
  namespace: dynatrace
spec:
  clusterIP: None
  selector:
    app: activegate
  ports:
    - port: 9999
      name: ag-https

```

Exposing ActiveGate for External Access

If external hosts need to route through the K8s ActiveGate:

```

# Option 1: LoadBalancer Service
apiVersion: v1
kind: Service
metadata:
  name: activegate-external
  namespace: dynatrace
  annotations:
    # AWS NLB
    service.beta.kubernetes.io/aws-load-balancer-type: "nlb"
    service.beta.kubernetes.io/aws-load-balancer-scheme: "internal"
spec:
  type: LoadBalancer
  selector:
    app: activegate
  ports:
    - port: 443
      targetPort: 9999
---
# Option 2: Ingress (with TLS passthrough)
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  name: activegate-ingress
  namespace: dynatrace
  annotations:
    nginx.ingress.kubernetes.io/ssl-passthrough: "true"
    nginx.ingress.kubernetes.io/backend-protocol: "HTTPS"
spec:
  ingressClassName: nginx
  rules:
    - host: activegate.internal.example.com
      http:
        paths:
          - path: /
            pathType: Prefix
            backend:
              service:
                name: activegate
                port:
                  number: 443

```


Platform-Specific Configurations

Amazon EKS

```
# EKS-specific annotations
apiVersion: v1
kind: Service
metadata:
  name: activegate
  namespace: dynatrace
  annotations:
    service.beta.kubernetes.io/aws-load-balancer-type: "nlb"
    service.beta.kubernetes.io/aws-load-balancer-internal: "true"
    service.beta.kubernetes.io/aws-load-balancer-cross-zone-load-balancing-
enabled: "true"
spec:
  type: LoadBalancer
  # ...
```

Azure AKS

```
# AKS-specific annotations
apiVersion: v1
kind: Service
metadata:
  name: activegate
  namespace: dynatrace
  annotations:
    service.beta.kubernetes.io/azure-load-balancer-internal: "true"
spec:
  type: LoadBalancer
  # ...
```

Google GKE

```
# GKE-specific annotations
apiVersion: v1
kind: Service
metadata:
  name: activegate
  namespace: dynatrace
  annotations:
    cloud.google.com/load-balancer-type: "Internal"
spec:
  type: LoadBalancer
  # ...
```

High Availability in Kubernetes

Configuration	Setting
Replicas	2-3 for production
Pod Anti-Affinity	Spread across nodes
PodDisruptionBudget	minAvailable: 1
Resource Requests	Guarantee scheduling

```
# Pod Anti-Affinity for HA
spec:
  affinity:
    podAntiAffinity:
      requiredDuringSchedulingIgnoredDuringExecution:
        - labelSelector:
            matchLabels:
              app: activegate
          topologyKey: kubernetes.io/hostname
---
# PodDisruptionBudget
apiVersion: policy/v1
kind: PodDisruptionBudget
metadata:
  name: activegate-pdb
  namespace: dynatrace
spec:
  minAvailable: 1
  selector:
    matchLabels:
      app: activegate
```

7. Verifying Deployment

After installation, verify ActiveGate is connected and healthy. ActiveGates surface in Grail as the **ACTIVEGATE Smartscape node**, so the queries below work directly in a notebook — no REST call needed.

A note on the classic path (ActiveGate 1.343, July 2026): ActiveGate 1.343 deprecates the classic `GET /api/v2/activeGates` endpoints in favor of this Smartscape node. Separately, ActiveGates have **never** been reachable through DQL `fetch` — there is no `dt.entity.active_gate` entity type in any spelling. Worth knowing precisely how that fails: `fetch dt.entity.active_gate` does not error — it returns **zero rows**, which is indistinguishable from "this environment has no ActiveGates." Treat an empty result from a classic entity fetch as a signal to check the entity type exists at all. `smartscapeNodes "ACTIVEGATE"` is the DQL path. The classic **Entities API v2** selector (`GET /api/v2/entities?entitySelector=type("ENVIRONMENT_ACTIVE_GATE")`) is a different surface from DQL and may still respond during the deprecation period — use it if you need a REST

fallback for a tenant that has not yet received 1.343, and migrate the automation on your next maintenance touch.

```
// ActiveGate inventory - every ActiveGate reporting to this environment,
// with its group, network zone, enabled modules, and host OS.
// Live-verified 07/30/2026. Costs nothing: Smartscape nodes scan 0 bytes of
// Grail.
smartscapeNodes "ACTIVEGATE"
| fields name, dt.active_gate.id, dt.active_gate.version,
dt.active_gate.group.name,
          dt.network_zone.id, is_containerized, os.type, os.version, modules
| sort name asc
```

```
// ActiveGate version spread - find the laggards before an update window.
// More than one row means the fleet is not uniform; sort ascending puts the
// oldest version first. Cross-check the OS end-of-support table in section 3
// against the os.version values from the inventory query above.
smartscapeNodes "ACTIVEGATE"
| summarize {activegates = count(), names = collectDistinct(name)},
  by:{dt.active_gate.version}
| sort dt.active_gate.version asc
```

```
// ActiveGate count per group and network zone - the HA check.
// Any row showing 1 is a single point of failure for that zone; production
// zones should show 2 or more (see High Availability in section 3).
smartscapeNodes "ACTIVEGATE"
| summarize activegates = count(), by:{dt.active_gate.group.name,
dt.network_zone.id}
| sort activegates desc
```

Verification via UI

Location: Deployment status → ActiveGates

Check for:

-  Status: Connected
-  Version: Latest or recent
-  Modules: Expected capabilities enabled

Verification Commands

Linux:

```
# Check service status
sudo systemctl status dynatracegateway

# Check connectivity
curl -k https://localhost:9999/communication/health

# View logs
sudo tail -100 /var/log/dynatrace/gateway/gateway.log
```

Windows:

```
# Check service status
Get-Service -Name "Dynatrace ActiveGate"

# Check connectivity
Invoke-WebRequest -Uri "https://localhost:9999/communication/health" -
SkipCertificateCheck
```

8. Troubleshooting

Common Issues

Issue	Cause	Solution
Not appearing in UI	Network blocked	Check firewall for 443 outbound
Shows disconnected	Service stopped	Restart dynatracegateway service
OneAgents not routing	Wrong network zone	Verify zone configuration
High memory usage	Too many agents	Add another AG or increase RAM
Certificate errors	Self-signed cert issue	Trust Dynatrace CA or use custom cert

Network Verification

```
# Test outbound to Dynatrace
curl -v https://{tenant-id}.live.dynatrace.com/communication/health

# Test ActiveGate is listening
netstat -tlnp | grep 9999

# Test from OneAgent host
curl -k https://{activegate-ip}:9999/communication/health
```

Log Locations

OS	Log Path
Linux	/var/log/dynatrace/gateway/
Windows	C:\ProgramData\dynatrace\gateway\log\

9. Next Steps

With ActiveGate deployed:

1. **ONBRD-04: Cloud & SaaS Integrations** — Connect AWS / Azure / GCP through your AG (or skip AG via the Clouds app where supported)
2. **ONBRD-05: Deploying OneAgent** — Now deploy agents that route through ActiveGate
3. Configure network zones if using multiple ActiveGates
4. Enable additional modules (synthetic, extensions) as needed
5. Set up monitoring for ActiveGate itself

Where to Go Deeper

- **K8S series** (15 notebooks) — Cluster monitoring, DynaKube depth, GitOps for K8s deployments
- **CLOUD series** (9 notebooks) — Per-cloud integration deep dives (AWS, Azure, GCP)
- **AUTOM series** — Configuration automation for ActiveGate deployment at scale

Deployment Checklist

Traditional (VM/Bare Metal)

- ☐ ActiveGate deployed in each required network zone
- ☐ At least 2 per zone for HA (production)
- ☐ Status shows "Connected" in Dynatrace
- ☐ Firewall rules configured (443 out, 9999 in)
- ☐ Network zones configured (if applicable)
- ☐ Sizing appropriate for expected OneAgent count

Kubernetes

- ☐ Dynatrace Operator installed (recommended)
- ☐ DynaKube CR applied with activeGate configuration
- ☐ Replicas set to 2+ for production HA
- ☐ Resource requests/limits configured
- ☐ PodDisruptionBudget created
- ☐ Pod anti-affinity configured for spread across nodes
- ☐ Service type appropriate (ClusterIP vs LoadBalancer)

Summary

In this notebook, you learned:

- What ActiveGate does and its capabilities
- When ActiveGate is required vs. optional
- Hardware requirements, the published per-shape capacity figures, and why FAQ-10 owns the sizing decision
- Which operating systems are currently supported, and which ones lose support inside the next six months
- Where to place ActiveGates in your network
- Installation methods for Linux, Windows, and containers
- **Kubernetes deployment** using Operator, Helm, or raw manifests

- Platform-specific configurations for EKS, AKS, and GKE
 - How to verify successful deployment with `smartscopeNodes "ACTIVEGATE"` inventory, version, and HA queries
-

References

General

- [ActiveGate Overview](#)
- [ActiveGate Installation](#)
- [Network Zones](#)
- [Linux ActiveGate hardware and system requirements](#)
- [Windows ActiveGate hardware and system requirements](#)
- [End-of-support announcements](#)
- [Private Synthetic Locations](#)
- [Clouds App](#)

Kubernetes

- [Dynatrace Operator](#)
 - [Dynatrace Operator GitHub](#)
 - [DynaKube parameters — ActiveGate configuration \(DT docs\)](#)
 - [DynaKube parameters \(DT docs\)](#)
 - [Helm Chart Repository](#)
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